Arenula

MCP server suite for the s&box game engine. Connects AI coding assistants to the editor in real time.

19
TOOLS

~120
ACTIONS

3
SERVERS

8-13K
TOKENS

ALL PHASES COMPLETE

Three Servers, One Suite

Editor

Scene manipulation, compilation, asset management, terrain editing. Runs inside the s&box editor as a C# plugin.

19 tools SSE on :8098 C#

API

Offline type and member reference. Indexes ~1,800 types and ~15,000 members from the s&box CDN schema.

8 tools stdio TypeScript

Docs

Narrative documentation fetcher. Scrapes docs.facepunch.com, converts to Markdown, caches with LRU.

3 tools stdio TypeScript

Editor Tools — 19 Omnibus Tools

Each tool uses an `action` parameter to select the operation. Grouped by resource, not operation type.

READ

scene

summary hierarchy statistics find find_in_radius get_details prefab_instances

7 actions

WRITE

gameobject

create destroy duplicate reparent rename enable set_tags set_transform batch_transform

9 actions

WRITE

component

add remove set_property set_enabled get_properties get_types copy

7 actions

NEW

compile

trigger status errors generate_solution wait

5 actions

WRITE

prefab

instantiate get_structure get_instances break update create save_overrides revert get_overrides

9 actions

READ

asset_query

browse | search | open | get_dependencies | get_model_info | get_material_properties | get_mesh_info | get_bounds | get_unsaved | get_status | get_json | get_references

12 actions

NEW

asset_manage

create | delete | rename | move | save | reload | get_references

7 actions

WRITE

editor

select | get_selected | set_selected | clear_selection | frame_selection | get_play_state | start_play | stop_play | get_log | save_scene | save_scene_as | undo | redo | console_list | console_run | open_code_file | get_preferences | set_preference

18 actions

WRITE

session

list | set_active | load_scene

3 actions

WRITE

lighting

create | configure | create_skybox | set_skybox

4 actions

WRITE

physics

add_collider configure_collider add_rigidbody create_model_physics create_character_controller create_joint

6 actions

WRITE

audio

create configure

2 actions

WRITE

effects

create configure_particle configure_post_processing

3 actions

WRITE

camera

create configure

2 actions

WRITE

mesh

create_block create_clutter set_face_material set_texture_params set_vertex_position set_vertex_color set_vertex_blend get_info

8 actions

NEW

navmesh

create_agent create_area create_link generate get_status query_path

6 actions

ASYNC

cloud

search get_package mount

3 actions

WRITE

project

get_collision set_collision_rule get_input get_info

4 actions

NEW

terrain

create configure get_info paint_material sync

5 actions

Architecture

```
AI Client ──> HTTP/SSE ──> ArenulaMcpServer (inside s&box editor)
                                  |
                            RpcDispatcher
                                  |
                  ┌───────────────┼───────────────┐
                  |               |               |
          async tools      console path    main thread
         (compile, cloud)  (exception       (all other
          background        isolation)       scene tools)
                  |               |               |
              Handler         Handler         Handler
```

28 C# files: 8 Core infrastructure + 20 Handlers (one per tool). Async tools dispatch before `GameTask.MainThread()` to avoid deadlocks.

Arenula vs Ozmium (Original)

METRIC	OZMIUM	ARENULA
Tools	105	**19**
Total actions	105	**~120**
Token cost (schemas)	~42,000	**~8-13,000**
MCP best practices	None	**Full Anthropic compliance**
Code compilation	Not available	**5 actions (trigger, status, errors, wait, generate_solution)**
Asset CRUD	Read-only	**Create, delete, rename, move, save**
Prefab authoring	Instantiate only	**Create, save overrides, revert, get overrides**
Terrain	None	**Create, configure, paint, sync**
NavMesh	Place only	**Generate, query paths, get status**
Cloud assets	Deadlocked	**Working (async dispatch)**
Error messages	Generic strings	**Structured with suggestions + fuzzy matching**
Pagination	Truncation only	**offset/limit with has_more metadata**
Response format	Raw dumps	**Canonical {id, name} tuples, concise/detailed modes**

METRIC	OZMIUM	ARENULA
Cross-platform	Windows DllImport	Clean (no P/Invoke)

Design Principles

Following Anthropic's official MCP guidance.

Omnibus Tools

Related operations grouped with `action` enum. Fewer tools = less selection ambiguity for the LLM.

Rich Descriptions

3-4 sentences per tool. What it does, when to use it, when NOT to use it, what actions are available.

Trimmed Responses

Canonical {id, name} tuples. Pagination with offset/limit. 16K char truncation. Concise/detailed modes.

Actionable Errors

Structured {error, action, suggestion} with fuzzy type matching. "Did you mean 'Rigidbody'?"

Schema Safety

`additionalProperties: false` on every schema. Enum constraints on action parameters. Prevents hallucination.

Async Safety

Network tools (compile, cloud) dispatch before MainThread. Never `.GetAwaiter().GetResult()` on main thread.

Installation

1. Copy the Editor plugin

```
# Copy into your s&box project's Libraries folder Arenula-MCP/editor/ →
YourProject/Libraries/arenula_mcp/
```

2. Build the API and Docs servers

```
# API server cd Arenula-MCP/api && npm install && npm run build # Docs server cd
Arenula-MCP/docs && npm install && npm run build
```

3. Configure your .mcp.json

```
{ "mcpServers": { "editor": { "type": "sse", "url": "http://localhost:8098/sse"
}, "api": { "command": "node", "args": ["path/to/Arenula-MCP/api/dist/index.js"]
}, "docs": { "command": "node", "args": ["path/to/Arenula-
MCP/docs/dist/index.js"] } } }
```

4. Open s&box

Arenula Editor compiles automatically and starts the MCP server on port 8098. Tools appear as `mcp__editor__*`, `mcp__api__*`, `mcp__docs__*`.

Based on Ozmium MCP Server by ozmium7 · API from sbox-api-mcp by sofianebel · Docs from sbox-docs-mcp

GPL-3.0 (Editor) · MIT (API, Docs)